



SESSION 01

FOUNDATIONS OF AUTONOMOUS SYSTEMS

Dynamical systems, state-space modeling, controllability, and observability.

ACTIVE TRACK
ALGORITHM-FIRST



MISSION CONTEXT

Robots are systems, not screenshots

Every robot evolves over time according to physics. Control theory describes, predicts, influences, and stabilizes that evolution. Before swarms, autonomy, or AI, we must understand one vehicle.

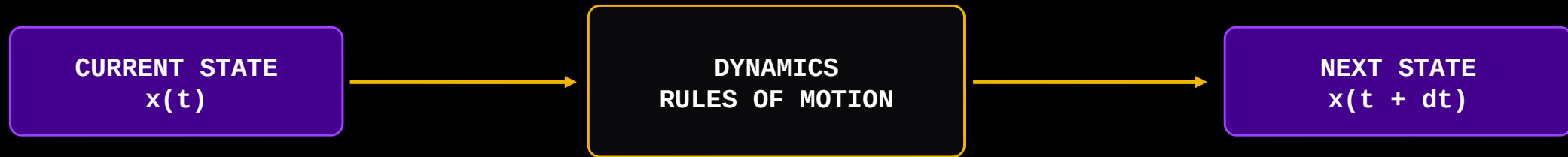


One vehicle first.
Many vehicles later.



CORE IDEA

A dynamical system changes state over time

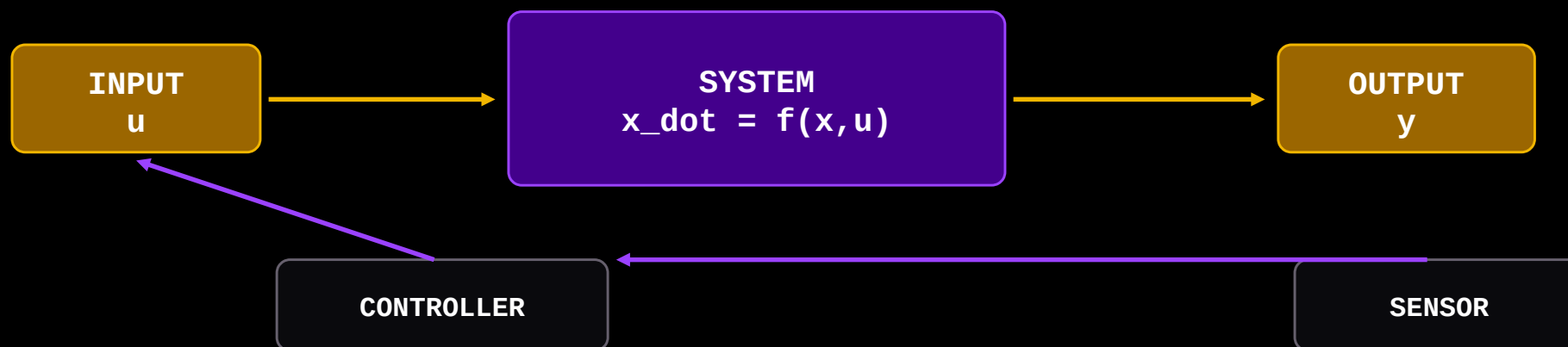


State = enough information to predict what happens next.



INPUT - STATE - OUTPUT

Control begins with information flow



Example:

u = propeller thrust

x = position + velocity + heading

y = GPS / compass / camera reading



MODEL CLASSES

Real robots are nonlinear. We approximate locally.

LINEAR MODEL

$$\dot{x} = A x + B u$$

Superposition works
Easier to analyze and design

NONLINEAR MODEL

$$\dot{x} = f(x, u)$$

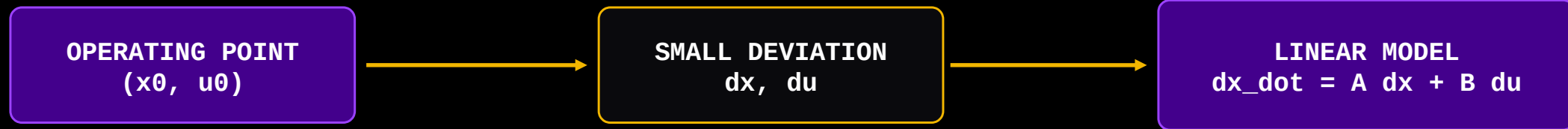
Drag, saturation, friction, geometry
Harder, but more realistic

Linear control is not denial of nonlinearity. It is a local engineering weapon.



WHY LINEARIZATION?

Make a hard system locally useful



For nonlinear dynamics:

$$\dot{x} = f(x, u)$$

Around (x_0, u_0) :

$$\delta \dot{x} = A \delta x + B \delta u$$

$$A = \left. \frac{df}{dx} \right|_{(x_0, u_0)}$$

$$B = \left. \frac{df}{du} \right|_{(x_0, u_0)}$$



Pendulum linearization

From $\sin(\theta)$ to θ

Nonlinear equation:

$$\ddot{\theta} + (g/L) \sin(\theta) = \tau / (m L^2)$$

This is nonlinear because $\sin(\theta)$ is nonlinear.

Small-angle assumption:

$$\begin{aligned}\theta &\approx 0 \\ \sin(\theta) &\approx \theta\end{aligned}$$

Valid when θ is small, e.g. < 10 - 15 degrees.

Physical meaning: near the hanging equilibrium, the pendulum behaves almost like a linear oscillator.



Pendulum: local linear model

Same system, simpler mathematics

Step 1: Start nonlinear
 $\theta_{ddot} + (g/L) \sin(\theta) = \tau / (m L^2)$

Step 2: Small angle
 $\sin(\theta) \approx \theta$

Step 3: Linearized equation
 $\theta_{ddot} + (g/L) \theta = \tau / (m L^2)$

State choice:

$x_1 = \theta$

$x_2 = \dot{\theta}$

Then:

$\dot{x}_1 = x_2$

$\dot{x}_2 = -(g/L)x_1 + (1/(mL^2)) \tau$

Takeaway: linearization turns nonlinear physics into state-space form.



Modern control language

Put dynamics into matrix form

Continuous-time LTI system:

$$\begin{aligned} \dot{x} &= A x + B u \\ y &= C x + D u \end{aligned}$$

x : internal state vector

u : control input vector

y : measured output vector

A, B, C, D : matrices defining dynamics and measurement

This form powers LQR, MPC, observers, Kalman filters, and swarm control.



What do A, B, C, D mean?

The four matrices are not decorative

A MATRIX

Natural system dynamics
How states affect each other

Example:
velocity changes position

B MATRIX

Input influence
How actuators push states

Example:
thruster changes velocity

C MATRIX

Sensor map
Which states are measured

Example:
GPS sees position

D MATRIX

Direct feedthrough
Input directly affects output

Often zero in robotics



Simple 1D marine robot

Position and velocity as state

State:

$x = [\text{position} ; \text{velocity}]$

Input:

$u = \text{thrust}$

Output:

$y = \text{position measurement}$

$$\dot{x} = A x + B u$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -b/m \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}$$

$$C = [1 \quad 0] \quad D = [0]$$

This is intentionally simple. The point is structure, not hydrodynamic perfection.



Can we move it?

Influence is the first control question



A system is controllable if suitable inputs can drive it from one state to another over time.



Matrix test, intuition first

Does input reach every state direction?

For n states:

$$Cmat = [B \quad A B \quad A^2 B \quad \dots \quad A^{(n-1)} B]$$

Controllable if:
 $\text{rank}(Cmat) = n$

Interpretation:

B tells where input enters.
 AB tells where input goes after dynamics.
 A^2B continues propagation.

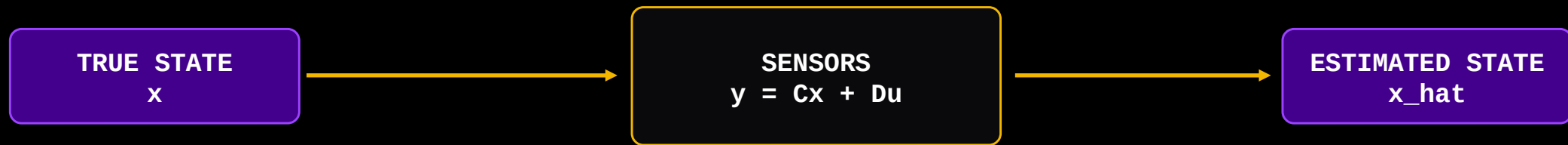
Full rank means input reaches all state directions.

Do not worship the rank test. Use it as a diagnostic: can the actuator influence the whole state?



Can we know it?

Information is the first estimation question



Observable means the hidden internal state can be reconstructed from measured outputs over time.



Matrix test, intuition first

Do measurements reveal every state direction?

For n states:

$$O_{\text{mat}} = \begin{bmatrix} C \\ C A \\ C A^2 \\ \vdots \\ C A^{(n-1)} \end{bmatrix}$$

Observable if $\text{rank}(O_{\text{mat}}) = n$

Interpretation:

C tells what the sensor directly sees.

CA tells what dynamics reveal next.

CA^2 keeps unfolding hidden state information.

Full rank means all states can be inferred.

Robotics translation: if you cannot measure it directly, can the motion history reveal it?



Two different questions

Influence and knowledge are not the same

CONTROLLABILITY

Question:

Can inputs move the state?

Matrix pair:

(A, B)

Failure:

Actuator cannot affect some mode

OBSERVABILITY

Question:

Can outputs reveal the state?

Matrix pair:

(A, C)

Failure:

Sensor cannot see some mode



Basic controller form

Use state to compute action



Closed-loop dynamics:

$$\dot{x} = (A - B K) x$$



What does K do?

It reshapes the closed-loop behavior

Open-loop:

$$\dot{x} = A x + B u$$

Feedback:

$$u = -Kx$$

Closed-loop:

$$\dot{x} = (A - BK)x$$

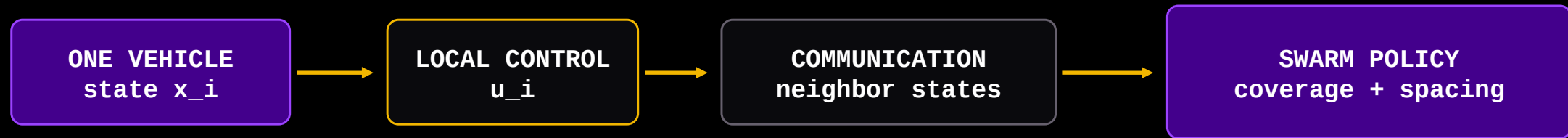
- Stabilize unstable motion
- Reduce tracking error
- Reject disturbances
- Trade speed against effort
- Prepare for LQR and MPC later

For today: K is not magic. It is a gain matrix that changes system behavior.



Single-agent control becomes swarm autonomy

Coordination layers sit above local control



Consensus intuition:

u_i depends on x_i and neighboring x_j

Local stability + coordination = useful swarm behavior



Where this fits in the company

Control theory is the spine of autonomy

Local robot control uses controllability.
State estimation uses observability.
Trajectory tracking uses feedback.
Formation control uses multi-agent dynamics.
Consensus uses state exchange between robots.

TECHNICAL LINEAGE

Simulation -> Control -> Estimation ->
Coordination -> Marine Swarms



Minimum vocabulary to remember

Enough to join future discussions

x: state
u: input
y: output
A: natural dynamics
B: actuator influence
C: sensor mapping
D: direct input-output path
K: feedback gain

Controllability: can we move it?
Observability: can we know it?
Linearization: local approximation
State feedback: $u = -Kx$
Closed loop: $\dot{x} = (A-BK)x$



What should survive after the lecture?

The core mental model

Every robot is a dynamical system.
State-space is the language of modern control.
Controllability asks whether inputs can move the system.
Observability asks whether sensors reveal the state.
Swarm autonomy is built on single-agent control.

If they remember only two lines:
Controllability = can we move it?
Observability = can we know it?



PEACOCK DYNAMICS

FROM SINGLE-AGENT CONTROL TO AUTONOMOUS MARINE SWARMS

The control foundation is now loaded. Next: LQR, MPC, estimation, and swarm coordination.

संप्रभुता । एकत्वम् । नियन्त्रणम् ।